GROUP WORK PROJECT #FE GWP 2 M5 MScFE 610: FINANCIAL ECONOMETRICS

GROUP NUMBER: 7681

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

Autocorrelation

Definition

Autocorrelation is used to measure the correlation of the time series with the similar time series shifted by k-lags. The autocorrelation at lag k, given a time series Xt, can also be expressed mathematically as:

$$\rho k = \frac{\sum_{t=k+1}^{T}(X_t - X')(X_{t-k} - X')}{\sum_{t=k+1}^{T}(X_t - X')^2}$$

Where:

ρk: Autocorrelation at lag k

X': Mean of the series X_t

T: Number of observations

Description

Autocorrelation is the correlation of a time series with its past or lagged values. If you are working on a time series, this is an important concept since it detects patterns, persistence, and dependencies in the data over time.

- Positive Autocorrelation: When the values of the series tend to be alike or in the same direction. In other words, in a stock or financial market, if the price of a stock is rallying today and continues to go up tomorrow, it is bullish momentum that reflects positive autocorrelation.
- Negative Autocorrelation: When values tend to go in opposite directions. For example, negative autocorrelation means that when the price of a stock goes up today, it will go down tomorrow.

Autocorrelation matters because it provides information about how the data is structured internally:

- Identifying Trend: Detects whether past values have an important impact on current values, which is crucial in predictive modelling.
- Pattern Recognition: Points out cyclical patterns or seasonality of time series data, for example, demand for products during holidays.

Nevertheless, autocorrelation stymies statistical modeling. For example:

- Autocorrelation challenges the assumption of independence in residuals in linear regression, which results in biased standard errors, ultimately risking the length hypothesis test.
- Ignoring autocorrelation in time series forecasting creates incorrect predictions.

Demonstration

In our example, we used daily stock returns from Nvidia to illustrate autocorrelation.

Durbin-Watson Statistic:
The test statistic checks for autocorrelation in residuals:

- Values close to 2 indicate no autocorrelation.
- Values less than 2 suggest positive autocorrelation.
- Values greater than 2 suggest negative autocorrelation.

Durbin-Watson Statistic: 1.9572

Differencing:
Removes trends or seasonality in time series data, making it stationary.
The first-order differenced returns are calculated by subtracting consecutive daily returns.

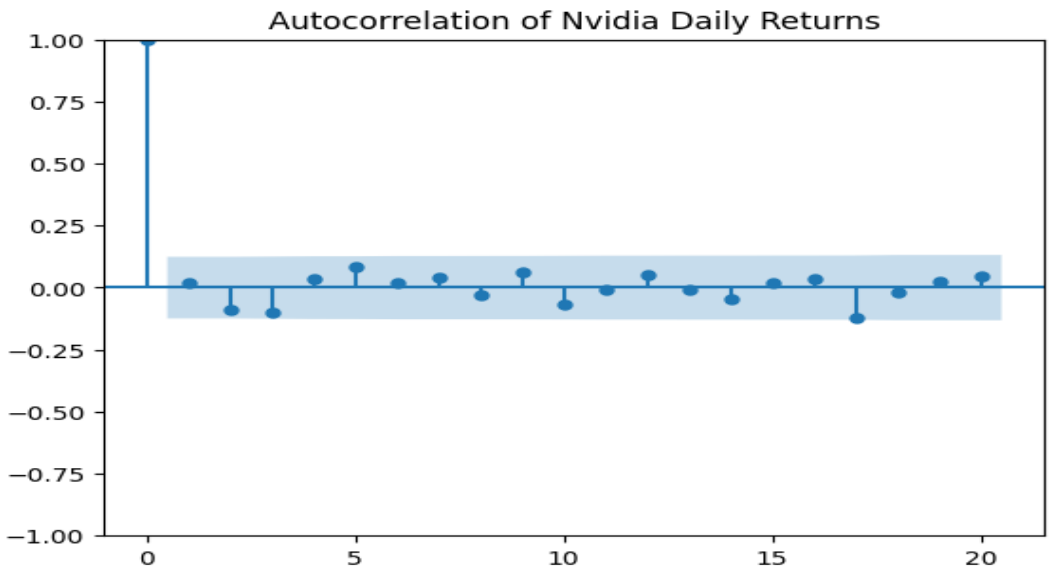
Autocorrelation Function (ACF):
Measures the correlation between a time series and its lagged values.
Used to assess the reduction in autocorrelation after differencing.

Model Summary:
Provides coefficients, significance tests, and fit statistics for the ARIMA model.
Includes metrics like AIC and BIC for model selection.

Residual Analysis:
Residuals represent the portion of the time series not explained by the model.
The ACF plot of residuals helps identify remaining patterns or autocorrelation.
Ideally, residuals should appear uncorrelated (white noise).

Diagram

We can also create an ACF plot to plot the autocorrelation of Nvidia's daily returns at lags. Lag 0 is 1 (it has a perfect correlation), which makes sense. All but the first few lags have a correlation around zero and fall within the confidence interval, so we have little or no significant autocorrelation. This also implies that the past returns of Nvidia are like a fair random walk, and this is in line with the efficient market hypothesis, where the past return has very little predictive power of future returns. The absence of notable patterns implies that more advanced modeling techniques may not gain the predictive advantage, as the data lacks memory and behaves independently over time.



Diagnosis

Tests and Indicators:

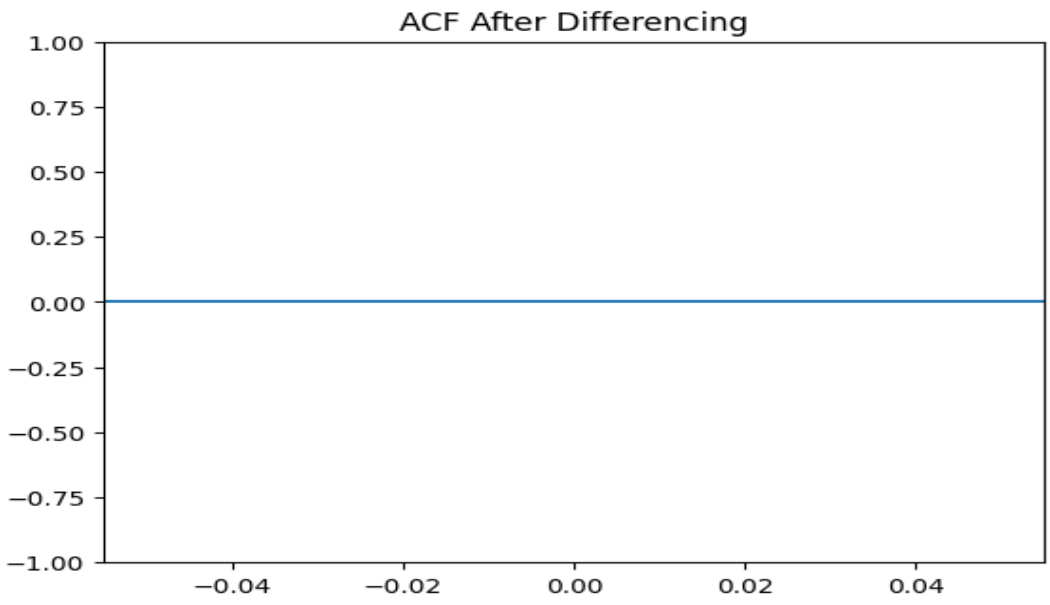
- Significant Peaks as depicted in Visual ACF Plot outside confidence bounds.
- Durbin-Watson Test: Tests for autocorrelation in residuals, e.g., in regression.

Damage

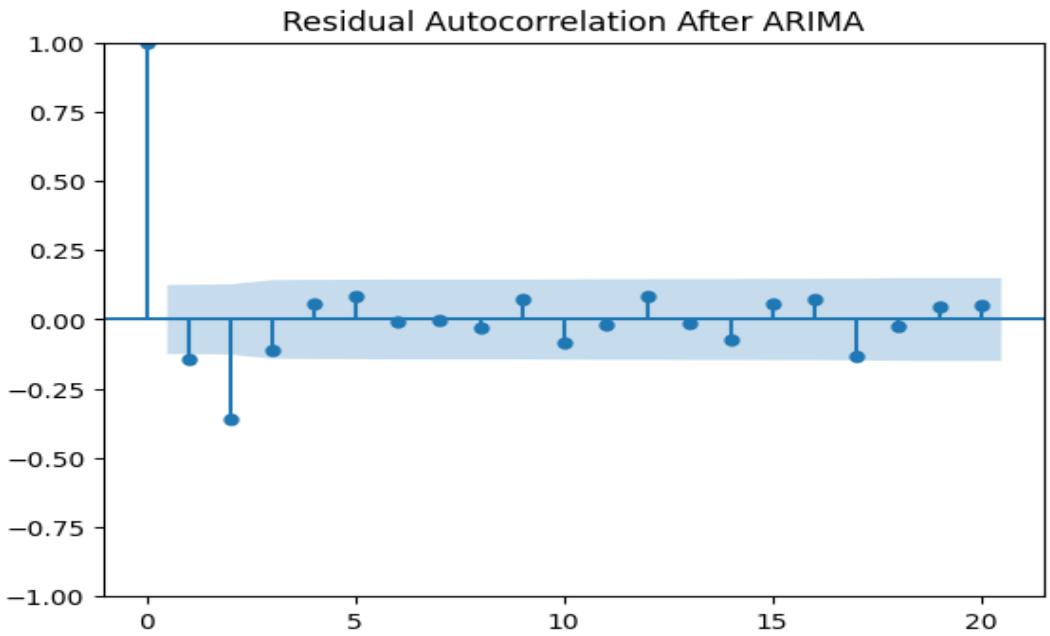
- Autocorrelation can skew forecasts and statistical inferences, resulting in the overestimation or underestimation of future values.
- The assumption of independence is critical in regression models.

Directions

1. Differencing: Removes autocorrelation by transforming the series: $X_t - X_{t-1}$.



2. ARIMA Models: Captures and models autocorrelation explicitly.



Heteroskedasticity

Definition

Heteroskedasticity is when the standard deviations of a predicted variable, monitored over different values of an independent variable or as related to prior time periods, are non-constant. (Hayes, 2024)

Description

Heteroskedasticity: In time series model the common structure of linear model is:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_n X_{in} + \epsilon_i$$

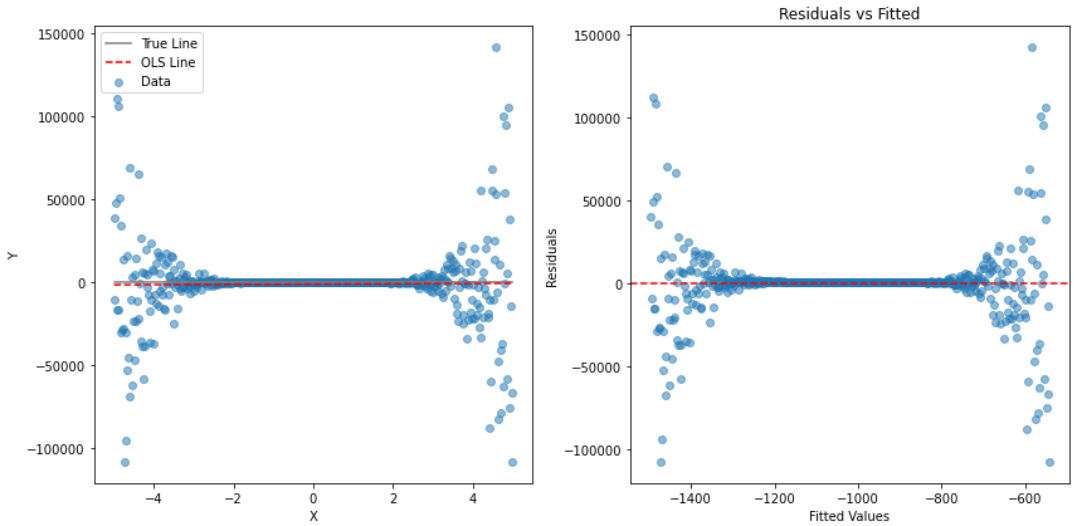
and it goes with common gauss - markov assumption where error has zero mean, and it has constant variance as well as uncorrelated and it is normally distributed.

When error has constant variance then we have what is called homoscedasticity and when the variance of error as well as residual term changes with time it is known as heteroskedasticity and it is given by Var(ϵi)=σ2∀i (IIT Kanpur and Shalabh)

Demonstration

Python code

Diagram



Diagnosis

The heteroskedasticity is found through several statistical tests available:

- Breusch Pagan test: This is done when variance is taken as a function of greater than one variable basically the explanatory variables.
- Goldfeld Quandt test: This is done when any one of the variables explain the heteroskedasticity.

- White test: This is done whether variance depends upon explanatory variables using cross product and their squares.

Damage

Heteroskedasticity leads to:

- Inefficient parameter estimates
- Incorrect risk assessments
- Unreliable Value at Risk (VaR) calculations
- Poor portfolio optimization decision Some of them have been shown in the code.

Directions

- ARCH family models for volatility clustering
- Robust standard errors for inference (Shalizi) (have been shown in the impact of heteroskedasticity python code)
- Rolling window volatility estimation implementation steps
- Test for ARCH effects',
- Select appropriate GARCH specification.
- Validate model assumptions.
- Implement volatility forecasting.

Over-reliance on normality

Definition

Normal and non-normal distributions are fundamental concepts in statistics that describe different patterns of data distribution. Normal distribution allows for straightforward statistical analysis and prediction. Non-normal distributions can exhibit various shapes and characteristics, making them more complex to analyze. The difference between normal and non-normal distributions is essential for analyzing data accurately. (geeksforgeeks) Assuming normal distribution without proper analysis can lead to poor statistical results and harm one's judgement.

Normal distribution probability density function:

$$f(X) = \frac{1}{\sigma_X \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{X-\mu_X}{\sigma_X}\right)^2}$$

Where $-\infty < X < \infty$

Description

To achieve correct assumptions, one should test normality on the data, to do so, we can turn to different methods, mainly graphs (visual) and statistical (numerical) tests.

The hypothesis in normality test:
H0: The data follows a normal distribution.
H1: The data does not follow a normal distribution.

Statistical Tests:

- Kolmogorov-Smirnov Test
- Shaprio-Wilk Test

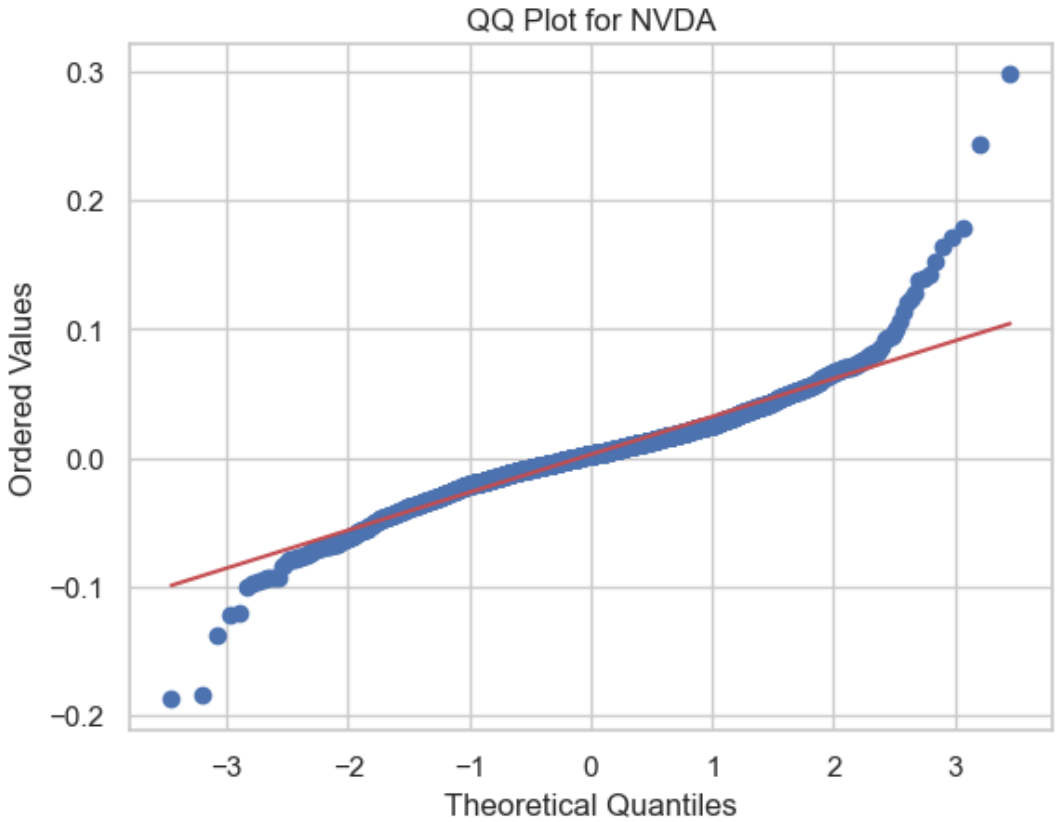
Graphs:

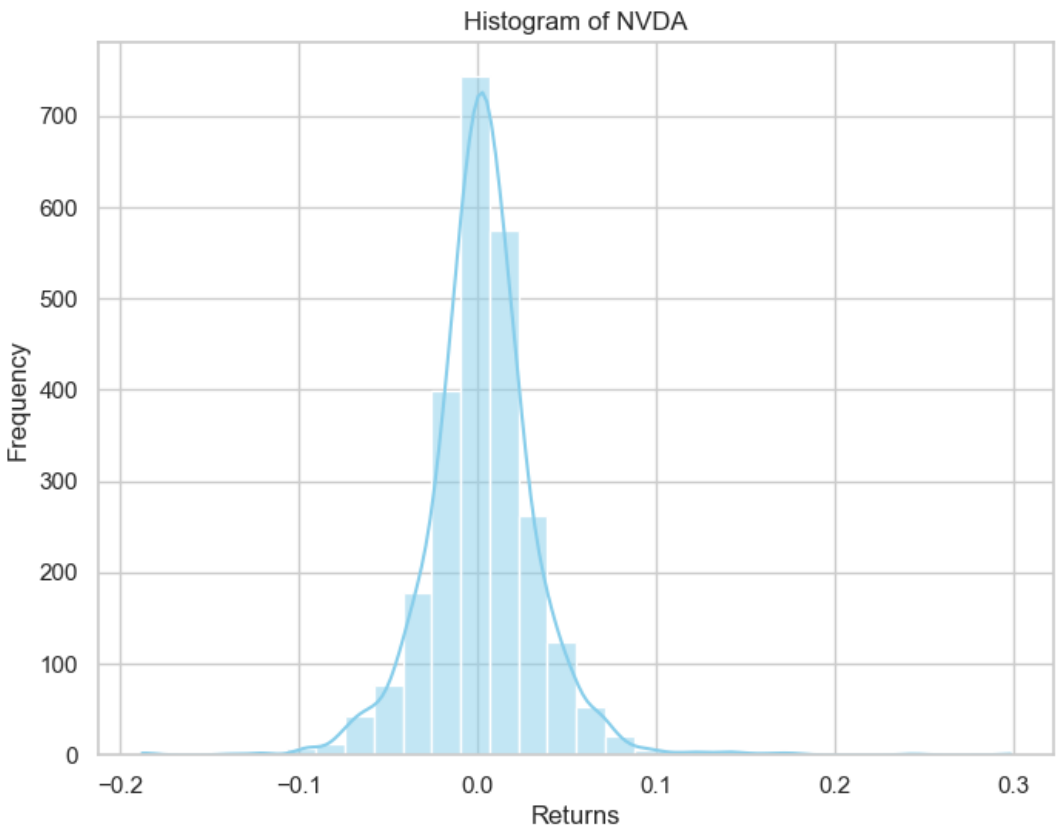
- QQ Plot
- Boxplot
- Distribution Histogram

Demonstration

Python code

Diagram


QQ Plot for NVDA
Ordered Values
Theoretical Quantiles
0.3
0.2
0.1
0.0
−0.1
−0.2
−3
−2
−1
0
1
2
3


Histogram of NVDA
Frequency
Returns
700
600
500
400
300
200
100
0
−0.2
−0.1
0.0
0.1
0.2
0.3

Diagnosis

The statistical tests and graph tests all lead to the same conclusion, that the returns are not normally distributed in our data.

The P-values are below 0.05 for both Shapiro-Wilk and Kolmogorov-Smirnov tests.

The QQ-plots suggest a heavy-tailed distribution whilst the histograms show the presence of positive kurtosis. Boxplots show the presence of many potential outliers.

Damage

Assuming normal distribution when the data is distributed differently can affect the:

- Confidence Intervals
- Mean and Standard Deviation
- Model Fit
- Predictions

Directions

Test for more suitable distributions to the data. Transform data with log returns or Box-Cox transformation to meet normality.

Heteroskedasticity and Non-normality Relationship

Damage

Both lead to unreliable risk measures

Directions

Use GARCH models with Student-t distributions.

Implement copula-based approaches.

Consider extreme value theory for tail risks.

Non-stationarity

Definition

A time series is said to be non-stationary if its statistical properties, such as mean, variance, and autocorrelation, vary with time. Non-stationarity is an important issue when modeling time series data, since many statistical models, including ARIMA, assume the data to be stationary.\ \ Stationarity Time Series: A time series Yt is stationary if for any point in time t, the following conditions are met (Baumöhl and Lyócsa): image.png

$$E(yt) = \mu, Var(yt) = \sigma^2, Cov(yt, yt + h) = \gamma h(constant)$$

Where:
μ is the mean,
σ^2 is the variance,
γh is the autocovariance at lag h.

Description

Non-stationarity often manifests in two ways:

1. Trend: A long-term upward or downward movement in the data.

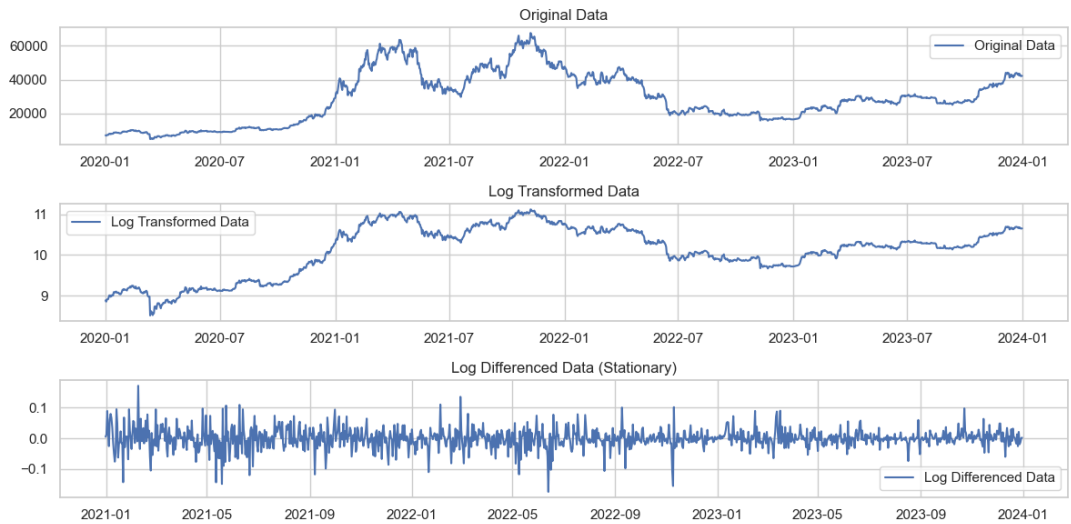
2. Seasonality: Periodic fluctuations at regular intervals.

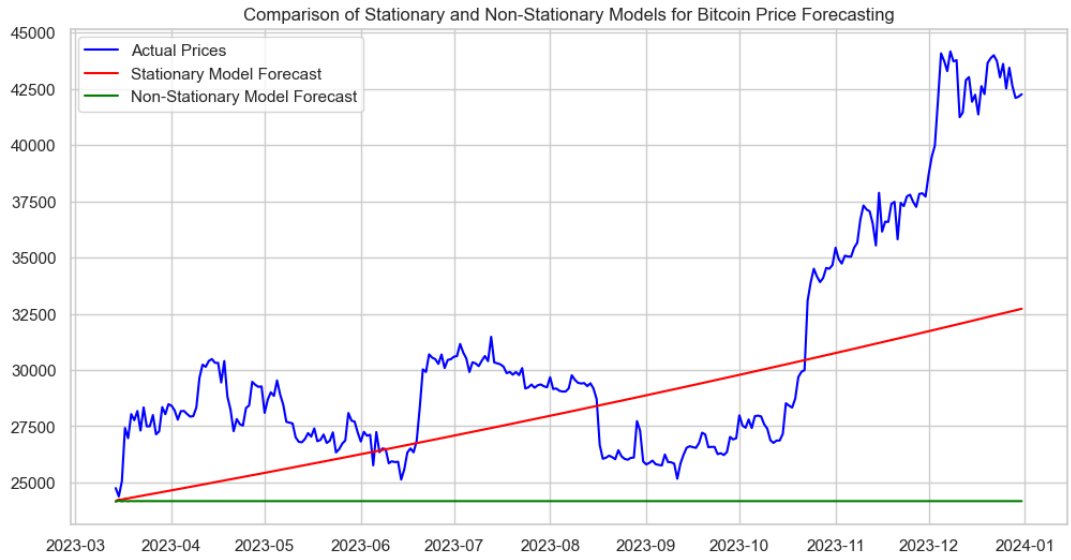
Non-stationarity due to, for example, trends, e.g., market growth, or volatility clustering is common in real-world financial data.

Demonstration

Python code

Diagram





Diagnosis

Visual Inspection: The rolling mean and standard deviation plots show both statistics changing significantly over time, indicating non-stationarity.

Statistical tests include Augmented Dickey-Fuller (ADF) Test. (Baumöhl and Lyócsa)

The ADF test checks if a time series has a unit root (non-stationary) If the test shows no unit root, the series is stationary.

If the test shows a unit root, the series is non-stationary and may need differencing.

The ADF test uses an augmented version of the standard Dickey-Fuller test, where lags of the differenced time series are included. This accounts for higher-order autocorrelations in the data.

p-value = 0.409 (> 0.05), indicating non-stationarity in the code

Damage

The damage caused by non-stationarity becomes clear in the forecasting results.

The non-stationary model (working with raw log-transformed data) performed significantly worse that is
MAE: 6,430.68 RMSE: 8,242.12

Whereas the stationary model (using log-differenced data) performed much better
MAE: 3,573.87 RMSE: 4,511.81

This means the non-stationary model's predictions were off by about 80% more on average!

Non-stationary financial data makes it extremely difficult to assess and manage risk properly. Traditional risk measures like Value at Risk (VaR) assume some level of stationarity in the underlying data. When this assumption is violated, risk models can severely underestimate potential losses.

Trading strategies that work well in one period might completely fail in another when dealing with non-stationary data. For example, if we are running an algorithmic trading system based on historical patterns, for buy n sell signals non-stationarity means these patterns might suddenly become invalid with false trading signals and portfolio loss.

If we see the Bitcoin data from our document. The wide swings in price and changing volatility patterns make it difficult for businesses to:

1. Plan cryptocurrency holdings
2. Set appropriate prices for crypto-based services.
3. Manage thei treasury operations effectively.

Directions

Log Transformations helps stabilize variance but doesn't completely solve the stationarity issue. Taking the difference between consecutive values removes trends. The code document has these two:

- Regular differencing (price[t] - price[t-1])
- Seasonal differencing (price[t] - price[t-365])

The successful approach which we used was log transformation followed by differencing, which:

- Stabilized the variance through log transformation and removed the trends though differencing and hence we got suitable stationary series for ARIMA modelling.

Other approaches could include GARCH volatility estimates, ARCH models are particularly good at capturing and forecasting volatility clustering - periods where big price changes tend to be followed by other big changes.

REFERENCES

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2. *IIT Kanpur, and Shalabh. "Heteroskedasticity."* https://home.iitk.ac.in/~shalab/econometrics/Chapter8-Econometrics-Heteroskedasticity.pdf
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5. *MscFE 610 Financial Econometrics*